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Technical Report

Issue Type Classification

Fine-Tuning BERT on BANKING77

AI System for Automated Customer Support Evaluation

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1 Dataset

We used **BANKING77** by Casanueva et al. (2020) — 13,083 customer support queries across 77 fine-grained banking intent categories.

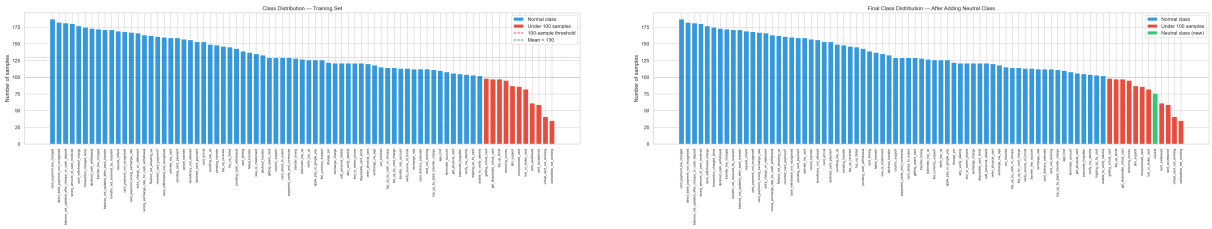
We extended it by adding a **neutral class** (95 sentences) for conversational filler phrases like *"Let me check that for you"*. These carry no issue-specific information and are excluded from voting during inference.

Table 1: Final dataset splits

Split	Samples	Classes
Train (original)	10,003	77
Test (original)	3,080	77
+ Neutral added	95	1
Train final	**10,079**	**78**
Test final	**3,092**	**78**

1.1 Class Distribution

The dataset has a **5.3× imbalance ratio** (max: 187 samples, min: 35 samples). We handled this with **inverse-frequency class weights** in the loss function, so the model penalizes minority class errors more.



(a) Original 77 classes

(b) After adding neutral class

Figure 1: Class distribution before and after adding the neutral class

2 Model and Fine-Tuning

We fine-tuned **bert-base-uncased** — 110M parameters, pretrained on Wikipedia and BooksCorpus, using the standard classification head (Dropout + Linear on the `[CLS]` token).

Table 2: Training hyperparameters

Parameter	Value	Reason
MAX_LEN	64 tokens	Avg sentence ∼15 tokens, covers 99%+
BATCH_SIZE	32	Stable gradients, fits GPU memory
EPOCHS	5	Sufficient for ∼10K samples
LR	2×10^{-5}	Standard BERT fine-tuning rate
Warmup	10%	Prevents instability at start
Optimizer	AdamW	Correct weight decay for transformers
Loss	Weighted CE	Handles 5.3× class imbalance
Early stop	Patience = 2	Restores best checkpoint automatically

Trained on **Kaggle** — Tesla T4 × 2 GPUs.

3 Results

3.1 Training Progress

Early stopping was not triggered — the model improved at every epoch.

Table 3: Epoch-by-epoch metrics

Epoch	Train Loss	Train Acc	Val Loss	Val F1
1	3.9862	12.18%	3.0637	0.4266
2	2.3975	63.68%	1.6569	0.7988
3	1.4159	84.00%	1.0409	0.8829
4	0.9543	90.27%	0.7898	0.9052
5	**0.7591**	**92.95%**	**0.7140**	**0.9130**

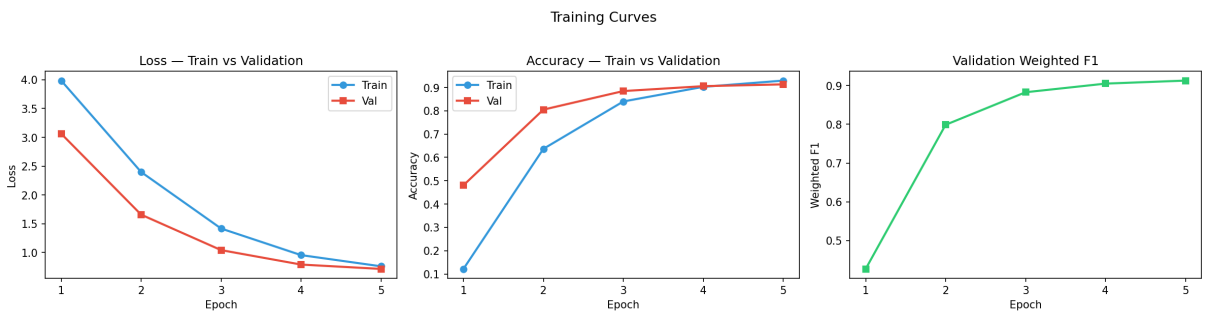


Figure 2: Loss, accuracy, and F1 per epoch. Train and validation curves stay close — healthy training with no overfitting.

3.2 Overfitting Analysis

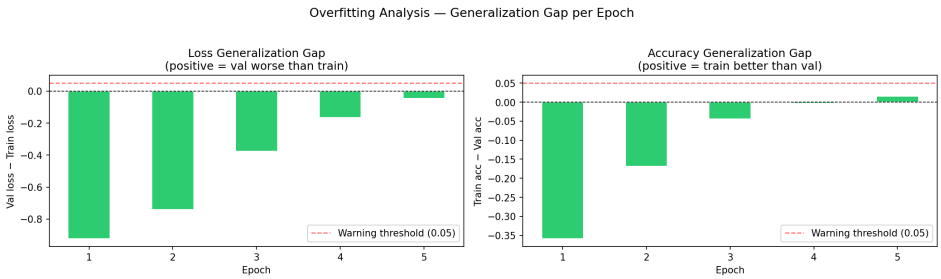


Figure 3: Generalization gap per epoch. Both gaps stay well below the 0.05 warning threshold.

Loss gap (val − train) = −0.0452 **Acc gap** (train − val) = +0.0161

Verdict: Good fit — no overfitting detected.

3.3 Final Metrics

Table 4: Final test set results

Metric	Value
Accuracy	**91.33%**
Weighted F1	**0.9130**
Macro F1	**0.9128**
Correctly classified	2,824 / 3,092
Classes below F1 = 0.70	1 out of 78

3.4 Per-Class F1 Score

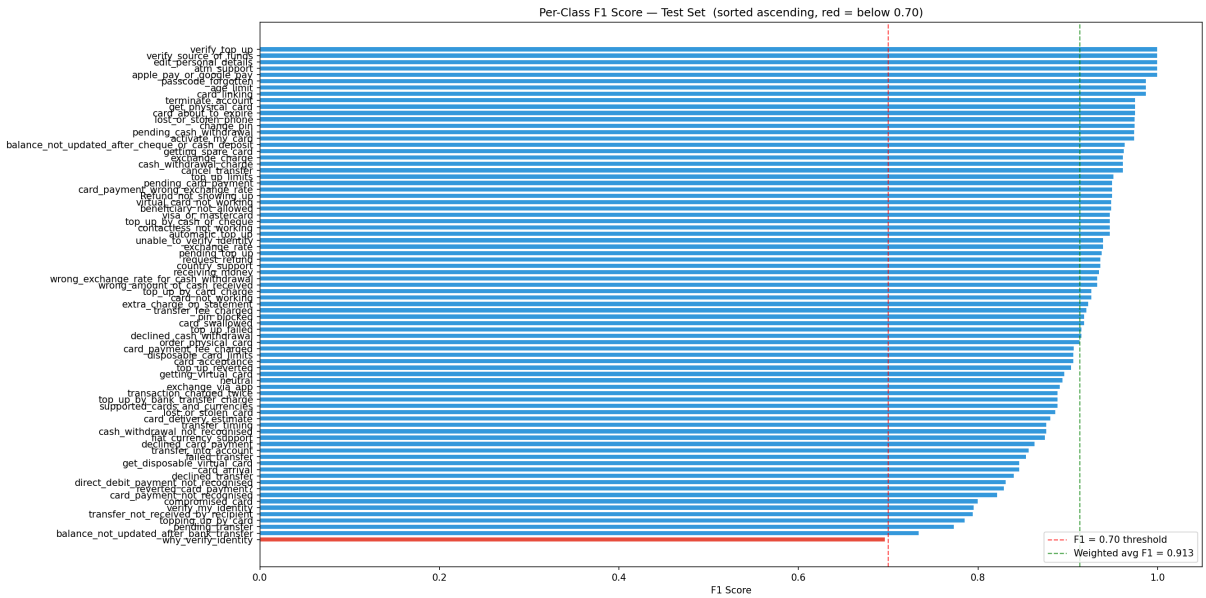


Figure 4: F1 score for all 78 classes sorted ascending. Only 1 class falls below the 0.70 threshold.

3.5 Confusion Matrix

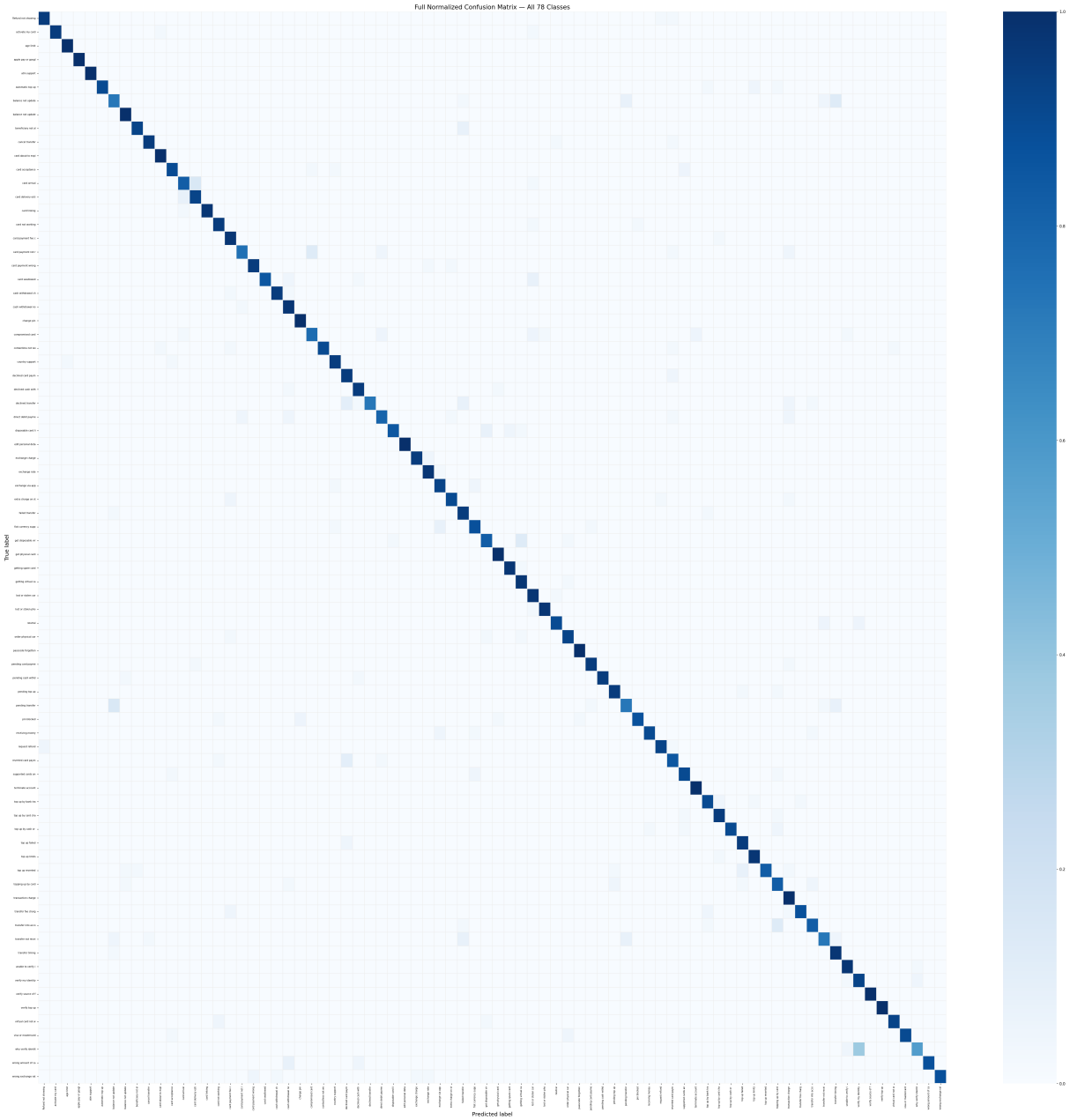


Figure 5: Full normalized confusion matrix — all 78 classes. A bright diagonal indicates consistent performance.

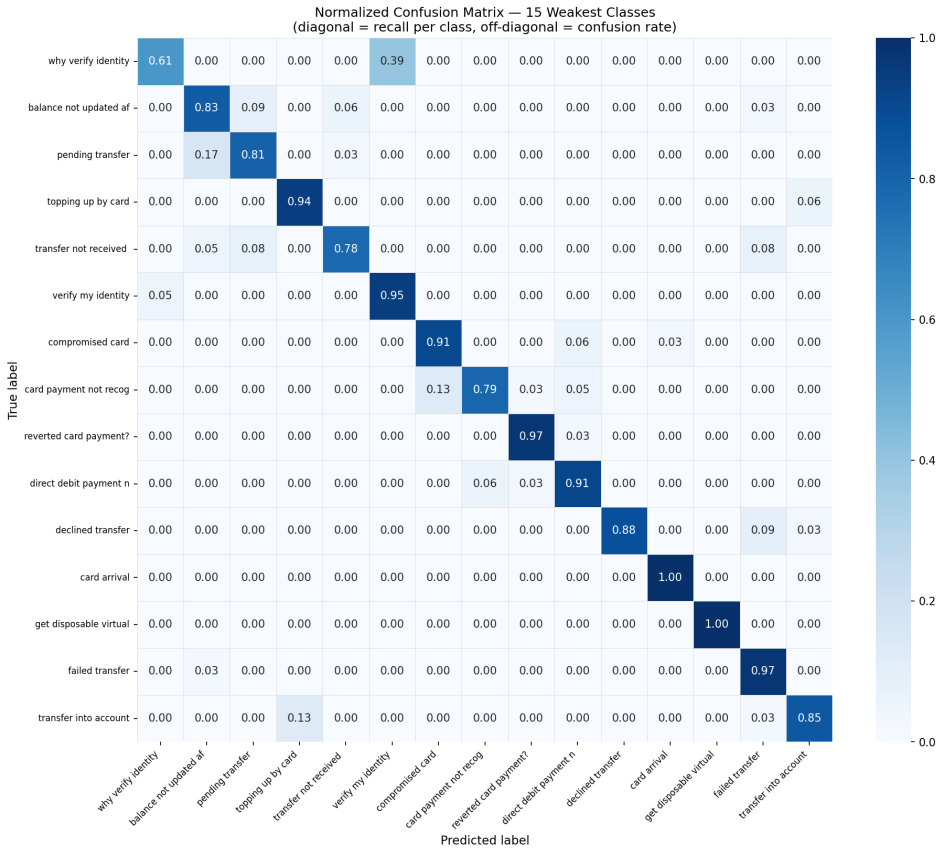


Figure 6: Focused matrix for the 15 weakest classes.

4 Inference Pipeline

The pipeline classifies a full call transcript at the sentence level and aggregates predictions via majority voting.

1. **Sentence splitting** — split on punctuation and newlines
2. **Per-sentence classification** — each sentence classified into one of 78 categories with a confidence score
3. **Neutral exclusion** — neutral-predicted sentences are excluded from voting
4. **Confidence filtering** — only sentences with confidence ≥ 0.60 vote
5. **Weighted majority voting** — vote weight = confidence score; highest total = primary issue
6. **Multi-label detection** — any label with $\geq 20\%$ vote share is also reported as a secondary issue

The model is hosted publicly at `Mohamed-Makram77/banking-issue-classifier` on Hugging Face Hub — no authentication required.